

E071

Applied Discrete Mathematics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A company with 600 employees has a rule: “If a user is an administrator, then the user can view audit logs.” It moved to a new ticketing system last year. Which statement must also be true?

- A. Every user can view audit logs.
- B. If a user cannot view audit logs, the user is not an administrator.
- C. If a user is not an administrator, the user cannot view audit logs.
- D. If a user can view audit logs, the user is an administrator.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

For the statement $p \rightarrow q$, which case makes it false?

- A. p is false and q is false.
- B. p is false and q is true.
- C. Any case where p is false.
- D. p is true and q is false.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A remote-access gateway, bought in 2022 from a large vendor, allows a session if (onVPN AND hasMFA). An engineer writes the deny branch as NOT(onVPN AND hasMFA). Which condition is equivalent?

- A. (NOT onVPN) OR (NOT hasMFA)
- B. onVPN OR hasMFA
- C. (NOT onVPN) AND hasMFA
- D. (NOT onVPN) AND (NOT hasMFA)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At a hospital with a 24-hour help desk, 120 staff have VPN access, 80 have admin access, and 30 have both. How many staff have at least one of these accesses?

- A. 140
- B. 200
- C. 170
- D. 30

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

200 users have at least one of two roles. 150 have Role A and 90 have Role B. How many have both?

- A. 60
- B. 440
- C. 50
- D. 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Each user account belongs to exactly one security group, and group membership is reviewed every quarter. Consider the relation “is in the same security group as” on user accounts. Which description is correct?

- A. A partial order.
- B. An equivalence relation: reflexive, symmetric, and transitive.
- C. Symmetric only.
- D. Not a relation, because membership changes each quarter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A legacy kiosk installed in 2009 accepts passwords of exactly 3 characters, each a lowercase letter or digit (36 choices), with repeats allowed. How many passwords are possible?

- A. 42,840
- B. 18,576
- C. 46,656
- D. 108

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A PIN uses digits 0–9 with repeats allowed. What is the shortest length that gives at least 1,000,000 possible PINs?

- A.** 6
- B.** 100,000
- C.** 1,000,000
- D.** 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A branch office links 6 routers in a full mesh so every router has a direct link to every other. The routers came from two different vendors. How many links are needed?

- A.** 30
- B.** 5
- C.** 15
- D.** 36

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A full-mesh network has 28 links. How many nodes does it have?

- A.** 8
- B.** 14
- C.** 7
- D.** 29

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A backup job written in Python by a contractor starts at 9 o'clock on a 12-hour clock and runs for 29 hours. What does the clock show when it ends?

- A.** 5 o'clock
- B.** 2 o'clock
- C.** 38 o'clock
- D.** 3 o'clock

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Under a Bell-LaPadula mandatory access control policy, a user with Secret clearance who has worked at the agency for 12 years tries to read a Top Secret file. The file's owner says it is fine. What happens?

- A. Allowed, because the owner approved it.
- B. Denied, because of the no-write-down rule.
- C. Allowed, because reading never changes data.
- D. Denied, because of the no-read-up rule.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A company ~~with 600 employees~~ has a rule: "If a user is an administrator, then the user can view audit logs." ~~It moved to a new ticketing system last year.~~ Which statement must also be true?

Target word: logically equivalent statement **Decides:** The contrapositive is equivalent.

A	Absolute language	The rule says nothing about non-admins.
B	Correct	Contrapositive $\neg q \rightarrow \neg p$.
C	Confusion set	The inverse, not equivalent.
D	Inverted operation	The converse; auditors might view logs too.

2. Answer D Concept 1 · transfer

For the statement $p \rightarrow q$, which case makes it false?

Target word: the one false row **Decides:** A promise is broken only when p holds and q does not.

A	Confusion set	Also vacuously true.
B	Inverted operation	Vacuously true.
C	Absolute language	A false p always makes the implication true.
D	Correct	The only false row of the truth table.

3. Answer A Concept 2

A remote-access gateway, ~~bought in 2022 from a large vendor,~~ allows a session if (onVPN AND hasMFA). An engineer writes the deny branch as NOT(onVPN AND hasMFA). Which condition is equivalent?

Target word: apply De Morgan **Decides:** $\neg(A \wedge B) = \neg A \vee \neg B$.

A	Correct	Negate both parts and flip AND to OR.
B	Inverted operation	Flips the connective but drops the negations.
C	Dropped element	Negates only one part.
D	Confusion set	Keeps AND; would allow sessions missing just one factor.

4. Answer C Concept 3

~~At a hospital with a 24-hour help desk,~~ 120 staff have VPN access, 80 have admin access, and 30 have both. How many staff have at least one of these accesses?

Target word: size of the union **Decides:** Subtract the overlap once.

A	Inverted operation	Subtracts the overlap twice.
B	Dropped element	Counts the 30 twice.
C	Correct	$120 + 80 - 30 = 170$.
D	Wrong stem number	The overlap only.

5. Answer D Concept 3 · transfer

200 users have at least one of two roles. 150 have Role A and 90 have Role B. How many have both?

Target word: solve for the intersection **Decides:** $|A \cap B| = |A| + |B| - |A \cup B|$.

A	Confusion set	$150 - 90$ is a difference, not an overlap.
B	Inverted operation	Adds all three instead of subtracting the union.
C	Dropped element	$200 - 150$ ignores Role B.
D	Correct	$150 + 90 - 200 = 40$.

6. Answer B Concept 4

Each user account belongs to exactly one security group, ~~and group membership is reviewed every quarter.~~ Consider the relation “is in the same security group as” on user accounts. Which description is correct?

Target word: classify the relation **Decides:** Check reflexive, symmetric, transitive.

A	Confusion set	Partial orders are antisymmetric; this relation is symmetric.
B	Correct	Each account shares its own group, the relation goes both ways, and it chains.
C	Dropped element	Misses reflexive and transitive, which also hold.
D	Camouflage	At any moment the relation is well defined.

7. Answer C Concept 5

A legacy kiosk ~~installed in 2009~~ accepts passwords of exactly 3 characters, each a lowercase letter or digit (36 choices), with repeats allowed. How many passwords are possible?

Target word: size of the password space **Decides:** 36 choices for each of 3 positions.

A	Confusion set	$36 \times 35 \times 34$ forbids repeats.
B	Adjacent category	$26^3 + 10^3$ counts all-letter or all-digit passwords only.
C	Correct	$36^3 = 46,656$.
D	Inverted operation	36×3 multiplies instead of raising to a power.

8. Answer A Concept 5 · transfer

A PIN uses digits 0–9 with repeats allowed. What is the shortest length that gives at least 1,000,000 possible PINs?

Target word: solve for length **Decides:** $10^n \geq 1,000,000$.

A	Correct	$10^6 = 1,000,000$.
B	Inverted operation	Divides 1,000,000 by 10 instead of taking a log.
C	Wrong stem number	The target count, not a length.
D	Dropped element	$10^5 = 100,000$ is too few.

9. Answer C Concept 6

A branch office links 6 routers in a full mesh so every router has a direct link to every other. ~~The routers came from two different vendors.~~ How many links are needed?

Target word: edges in K_6 **Decides:** $n(n - 1)/2$ with $n = 6$.

A	Dropped element	Forgets to divide by 2, counting each link twice.
B	Adjacent category	Links for a star topology ($n - 1$).
C	Correct	$6 \times 5 / 2 = 15$.
D	Confusion set	n^2 counts self-links and duplicates.

10. Answer A Concept 6 • transfer

A full-mesh network has 28 links. How many nodes does it have?

Target word: solve for n **Decides:** $n(n - 1)/2 = 28$.

A	Correct	$8 \times 7 / 2 = 28$.
B	Inverted operation	Divides 28 by 2.
C	Dropped element	That is $n - 1$.
D	Adjacent category	Nodes for a star with 28 links.

11. Answer B Concept 7

A backup job ~~written in Python by a contractor~~ starts at 9 o'clock on a 12-hour clock and runs for 29 hours. What does the clock show when it ends?

Target word: wrap-around time **Decides:** $(9 + 29) \bmod 12$.

A	Adjacent category	$29 \bmod 12$ ignores the 9 o'clock start.
B	Correct	$38 \bmod 12 = 2$.
C	Dropped element	Does not reduce mod 12.
D	Confusion set	$38 \div 12$ has quotient 3; the quotient is not the answer.

12. Answer D Concept 8

Under a Bell-LaPadula mandatory access control policy, a user with Secret clearance ~~who has worked at the agency for 12 years~~ tries to read a Top Secret file. The file's owner says it is fine. What happens?

Target word: MAC read decision **Decides:** The file is above the user's clearance.

A	Confusion set	Owner approval is discretionary control; MAC labels override it.
B	Inverted operation	Right outcome, wrong rule: this is a read, and the file is higher.
C	Dropped element	Ignores that confidentiality limits reading.
D	Correct	Subjects cannot read above their clearance.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Implication and its contrapositive	$p \rightarrow q$ is false only when p is true and q is false; it is equivalent to its contrapositive ($\neg q \rightarrow \neg p$), not its converse or inverse.	Treating the converse ($q \rightarrow p$) as equivalent.	An if-then policy and “which must also be true?”
2	De Morgan’s laws	$\neg(A \wedge B) = \neg A \vee \neg B$ and $\neg(A \vee B) = \neg A \wedge \neg B$: negate each part and flip AND/OR.	Negating each part but keeping the same connective.	“Deny unless both...” or the NOT of a compound condition.
3	Inclusion-exclusion	$ A \cup B = A + B - A \cap B $.	Adding the two sets and double-counting the overlap.	Two overlapping groups and “at least one.”
4	Equivalence relations	A relation that is reflexive, symmetric, and transitive is an equivalence relation and splits a set into non-overlapping classes.	Checking only one property, or confusing it with a partial order (reflexive, antisymmetric, transitive).	“Is in the same group as” on a set of items.
5	Counting a password space	With repetition allowed, the number of strings is $(\text{characters available})^{\text{length}}$.	Multiplying characters by length, or using a no-repeat permutation.	Character set size and a fixed length.
6	Edges in a complete graph (full mesh)	A complete graph on n nodes has $n(n - 1)/2$ edges; the degree sum always equals twice the number of edges.	Forgetting to divide by 2, or using n^2 .	“Every device linked directly to every other.”
7	Modular arithmetic	$a \bmod n$ is the remainder after dividing by n ; clocks, cyclic shifts, and many ciphers wrap around this way.	Reporting the quotient or the unreduced sum.	Wrap-around: clocks, days of the week, alphabet shifts.
8	Mandatory access control lattice (Bell-LaPadula)	For confidentiality, a subject may not read objects above its clearance (no read up) and may not write to objects below it (no write down); the system, not the owner, enforces labels.	Letting an owner’s grant override the labels, as in discretionary control.	Clearance levels and labeled files.

Course scope (public description): Applied Discrete Mathematics introduces the foundational mathematical structures that underpin modern technology. Topics include logic, Boolean algebra, sets, functions, relations, graphs, combinatorics, and modular arithmetic, with emphasis on their applications to technology concepts such as Boolean algebra and structures, password complexity, access control models, network topologies, and cryptography. Students will not focus on abstract proofs but will instead learn how discrete mathematics informs real-world technology structures and practices. By the end of the course, students will be able to apply mathematical reasoning to technology scenarios, analyze system vulnerabilities through discrete models, and explain the mathematical principles behind technological architectures and protocols.